9173
H2 Geography Paper 1 Suggested Pointers
TJC Preliminary Examination

Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

1

(a) Using Resource 1, describe the pattern of progress towards SDG7. [4]

Possible responses include:
- Majority of countries in the world have not achieved SDG7
- Majority of countries which have achieved SDG7 are developed countries
- Worst performing continent is Africa
- There are no countries in Asia which have achieved SDG7

(b) Explain possible challenges limiting the progress of countries towards achieving SDG7 as shown in Resource 1. [6]

Several challenges limit the progress of countries toward achieving Sustainable Development Goal 7 (SDG7), which aims to ensure access to affordable, reliable, sustainable, and modern energy for all. These challenges vary across regions and are influenced by socio-economic, political, environmental, and technological factors. Here are some of the key obstacles:

- Many developing countries, especially in rural areas, lack the infrastructure to generate, distribute, and maintain energy supplies. The cost of building energy grids and maintaining transmission lines in remote areas is high, making it difficult to extend access to affordable energy.
- Aging infrastructure in some developed countries also limits energy efficiency and the adoption of renewable energy sources.
- High investment costs: Transitioning to clean energy requires substantial upfront investments in technology, infrastructure, and research. Many countries struggle with limited financial resources, inadequate access to funding, or reliance on external financing, which may come with conditionalities.
- Energy poverty: Low-income households may not be able to afford modern energy services, and countries with large populations living in poverty often face difficulties balancing affordability with sustainable energy provision.
- Many countries, especially those rich in fossil fuel resources (e.g., coal, oil, and natural gas), are economically dependent on fossil fuel industries. Transitioning to clean energy threatens jobs and revenue streams in these sectors, leading to resistance from political and business interests.
- Subsidies for fossil fuels in some countries discourage the adoption of renewable energy sources, making fossil fuels artificially cheaper and harder to phase out.
- Inconsistent or weak government policies regarding clean energy transition hinder progress. Some governments lack clear long-term plans, or policies may change with shifts in political leadership.
- Bureaucratic hurdles, corruption, and lack of accountability can slow the implementation of renewable energy projects

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- Limited access to technology: Many developing countries struggle with limited access to the latest clean energy technologies and expertise, impeding their ability to implement solutions like solar, wind, or energy storage systems.
- Slow innovation adoption: Even in countries where technology is available, the adoption of clean energy innovations can be slow due to market entry barriers, lack of skilled labor, or inadequate research and development (R&D).
- Renewable energy sources like solar and wind are intermittent, meaning energy generation depends on weather conditions. The lack of reliable energy storage solutions or grid flexibility makes it challenging to ensure a continuous energy supply.
- High costs of energy storage technologies, such as batteries, and the need for grid modernization to integrate renewable sources add complexity to the transition.
- Energy supply chains, especially for clean energy technologies, rely on rare earth materials, the extraction and production of which are concentrated in a few countries. Geopolitical tensions and resource competition can disrupt access to these essential materials.
- Conflicts, political instability, and insecurity in some regions make it difficult to develop or maintain energy infrastructure.
- Large-scale renewable energy projects, such as hydropower dams, can have significant environmental impacts, such as displacing communities or harming ecosystems. This can lead to public opposition or legal battles, delaying progress.
- In some cases, there is a lack of community engagement and social inclusion in energy projects, leading to inequities in energy access and conflicts over resource use
- Many countries lack the institutional capacity to plan, implement, and regulate complex energy systems. Weak governance, limited regulatory frameworks, and insufficient enforcement of energy policies impede progress.
- Poor coordination among stakeholders (governments, private sector, civil society) also slows down energy initiatives.
- Climate change affects energy systems by increasing the frequency of extreme weather events (e.g., floods, droughts) that can damage energy infrastructure or reduce the efficiency of certain energy sources, such as hydropower.
- Adaptation to climate-related risks requires additional investment in making energy infrastructure more resilient, which adds to the cost of transitioning to clean energy.

(c) Citing data from Resource 2, compare the energy sources for Germany between 2013 and 2023. [4]

Possible responses include:
- The number of energy sources reduced
- In 2013 the main energy source was nuclear power
- The energy source with the greatest increase from 2013 to 2023 is natural gas
- renewable energy sources in total energy mix doubled

(d) Using Resource 3, explain possible trade-offs Gambia will need to consider should the proposed dam and HEP station be constructed. [6]

Several trade-offs will need to be considered, as such projects have both potential benefits and significant socio-economic, environmental, and geopolitical impacts. The decision-making process will require weighing the advantages of improved energy access against the costs associated with various factors. Below are some key trade-offs:

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- Environmental impacts include the potential for habitat destruction, changes in river ecosystems, and reduced biodiversity. Flooding from the dam's reservoir could submerge forests, agricultural land, or even entire communities. Fish migration patterns and aquatic life could also be disrupted, harming local wildlife and fisheries.
- The project may lead to social displacement of people living near the proposed dam site, forcing them to relocate. This can result in loss of homes, land, and livelihoods, particularly for communities that rely on agriculture or fishing. Compensation schemes may not fully address the socio-cultural impacts of displacement.
- The construction of a dam may lead to reduced water flow downstream, negatively impacting food security. If water availability decreases, agricultural land downstream may suffer from drought or reduced irrigation capacity, harming crop yields. Fishing, a key source of food and income in river communities, could also be adversely affected by changes in water flow or fish migration.
- Ecosystem services provided by the river, such as water purification, flood regulation, and support for biodiversity, could be disrupted by the dam. The long-term environmental consequences of altering the river's natural flow may outweigh the short-term benefits of cleaner energy.
- The upfront costs of dam construction can be extremely high, especially for a small economy like The Gambia. Financing such a project may require international loans, which could lead to increased debt. Long construction timelines may delay economic benefits, and cost overruns are a common risk in large infrastructure projects.
- The dam could create geopolitical tensions with neighboring countries that share the river basin. Since The Gambia relies on the Gambia River, which flows through multiple countries (Guinea, Senegal, etc.), altering its flow could lead to disputes over water rights. Reduced water flow downstream could provoke conflicts, particularly with countries like Senegal, which also depend on the river for agriculture and energy.
- Climate change could pose risks to the dam itself. Dams rely on consistent water flows, but climate-induced changes such as altered rainfall patterns, prolonged droughts, or extreme weather events (e.g., floods) could make the energy generation from hydropower less reliable. This could undermine the project’s long-term sustainability and resilience.
- Local livelihoods, particularly those dependent on agriculture, fishing, or tourism around the river, may suffer. Disruption of ecosystems or reduction in fish populations could harm those relying on fishing, while agricultural productivity may decrease due to altered water access. Local communities may not benefit equally from the economic growth generated by the dam.

(e) Study Resource 4. Explain the potential and limitations of solar energy in providing energy security to Singapore. [6]

Possible responses for the potential of solar energy include:
- Reliability of energy supply at reasonable prices
- The potential to harness solar power based on Singapore’s location
- Singapore as DC and ability to R&D

Possible responses for limitations of solar energy include:
- Lower efficiency of solar
- Space demands not met

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- Current weather and environmental conditions
- Cost issues

(f) Suggest reasons to explain why extractive industries are often dominated by large private and state-owned firms. [4]

Extractive industries, which include sectors like mining, oil, gas, and other natural resource extraction, are often dominated by large private and state-owned firms for several key reasons. These reasons reflect the capital intensity, technical complexity, and geopolitical significance of the extractive sector, which tends to favor larger organizations with the necessary resources, expertise, and governmental backing. Here's why these industries are dominated by such large firms:

- massive upfront investments in exploration, extraction, transportation, and processing of natural resources.
- technological capabilities and skilled workforce to operate complex extraction machinery, manage large-scale operations, and process raw materials efficiently
- economies of scale
- risk management ability

Cluster 2: Tropical Environments

2

(a) Using Resource 5, describe the distribution of karst in tropical China and Southeast Asia. [4]

Possible responses include:
- formed near large river systems
- Discontinuous patches
- distinct patch that runs parallel to each othe
- presence of karst off the coast

(b) Compare the features of the two karst landforms shown in Resource 6. [6]

Possible responses include:
- type
- form
- height shape
- presence/absence of vegetation
- steepness
- surrounding environ

(c) Explain the formation of the speleothems shown in Resource 7. [6]

Speleothems are secondary mineral deposits formed within caves, primarily composed of calcium carbonate, which accumulate over long periods. Their formation begins with the interaction between water and carbon dioxide in the atmosphere and soil. As rainwater passes through the soil, it absorbs carbon dioxide, forming weak carbonic acid. When this slightly acidic water infiltrates through cracks and fissures in limestone or other soluble rocks, it dissolves small amounts of calcium carbonate from the surrounding rock, carrying it in solution.

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As the water continues to seep into a cave, the environment inside, typically less saturated with carbon dioxide than the outside, causes the water to lose some of its carbon dioxide through evaporation or a decrease in pressure. This loss of carbon dioxide reduces the water's ability to hold calcium carbonate in solution, leading to the precipitation of the mineral. Over time, the gradual deposition of calcium carbonate forms speleothems.

Different types of speleothems develop depending on the location of the dripping water and the rate of deposition. Stalactites form when water drips from the ceiling of a cave, leaving deposits behind as it evaporates. The accumulation grows downwards, creating elongated, icicle-like formations. Conversely, stalagmites are formed when water drips onto the cave floor, and the calcium carbonate is deposited in upward-growing mounds. If a stalactite and stalagmite meet, they can form a column. Additionally, flowstones are formed when water flows over walls or floors, depositing thin layers of minerals in a smooth, sheet-like formation.

(d) Suggest two ways in which human activities can negatively impact ecosystem services provided by karst landforms. [4]

Possible responses include:
A threat to karst ecosystems is pollution, particularly from agricultural runoff, industrial waste, and sewage. Karst systems have porous rock formations and extensive underground drainage networks, which allow pollutants to travel quickly through the system with little filtration. This can lead to contamination of underground rivers and aquifers, which many communities rely on for clean drinking water. Agricultural chemicals such as fertilizers, pesticides, and animal waste can seep into the groundwater, harming both water quality and the biodiversity that thrives in subterranean karst environments.

The development of urban areas, roads, and infrastructure on or near karst landscapes can cause extensive damage. Construction activities often lead to the clearing of vegetation, which plays a crucial role in stabilizing the soil and regulating water infiltration. Without vegetation, soil erosion increases, and more surface water flows into the karst system, potentially causing floods, the collapse of sinkholes, or overloading underground drainage. Urbanization can also increase impervious surfaces (like concrete), reducing natural water infiltration and disrupting the groundwater recharge process that karst systems are integral to. As water quality deteriorates and water flows are altered, the ecosystem services provided by karst landscapes—such as flood control and groundwater storage—are diminished.

Karst areas are often rich in limestone and other minerals, leading to extensive mining and quarrying activities. These industries physically alter the landscape, removing rock and damaging surface and subsurface formations like caves and underground rivers. Mining can disrupt the natural drainage patterns, lower the water table, and cause subsidence or sinkholes. Quarrying can also release pollutants into the groundwater, further degrading water quality and threatening the organisms and ecological systems dependent on clean, stable environments. The destruction of these natural features reduces the ability of karst landscapes to provide water purification and habitat services.

While karst landscapes, especially caves, attract tourism due to their scenic beauty and unique geological features, unmanaged or poorly regulated tourism can lead

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to significant environmental degradation. Large numbers of visitors can disturb fragile cave environments, damaging speleothems (mineral formations like stalactites and stalagmites) and disturbing the delicate ecological balance of cave ecosystems. Tourism infrastructure, such as trails, parking lots, and accommodations, can also contribute to habitat fragmentation, vegetation loss, and soil erosion. Additionally, waste generated by tourists, if not properly managed, can pollute water systems and harm local wildlife.

In many karst regions, forests and other natural vegetation are cleared for agriculture or logging. Deforestation significantly disrupts the natural hydrological cycle, as trees and plants are essential for regulating water infiltration and maintaining soil stability. The removal of vegetation leads to increased surface runoff, which can cause flooding, accelerate soil erosion, and contribute to the collapse of karst features like sinkholes. Intensive agriculture, particularly in karst regions, can also lead to the overuse of water resources for irrigation, depleting groundwater levels. Fertilizer and pesticide runoff from farming activities pollutes underground water systems, harming both the natural environment and the people who rely on these aquifers for drinking water.

Human-induced climate change can have indirect but severe impacts on karst ecosystems. As global temperatures rise and weather patterns become more erratic, karst landscapes may experience more extreme rainfall events, which can lead to increased flooding and erosion. Conversely, in some regions, climate change may reduce rainfall, leading to drought and the drying up of groundwater reserves. Changes in precipitation and temperature can also disrupt the ecosystems that depend on karst environments, leading to biodiversity loss. The capacity of karst landscapes to provide essential services like water storage, climate regulation, and carbon sequestration can be significantly diminished under changing climate conditions.

Karst aquifers are crucial sources of fresh water, but over-extraction of groundwater for agriculture, industry, or municipal use can deplete these reserves and disrupt the natural balance of the system. When too much water is drawn from a karst aquifer, the water table can drop, leading to the drying out of springs and underground rivers, the collapse of caves, and the formation of new sinkholes. This can reduce the ability of the karst system to provide clean, reliable water supplies and harm the species that rely on karst wetlands and underground habitats.

(e) Compare the relationship between vertical drop and horizontal travel distance of the two mass movement processes shown in Resource 8. [4]

Possible responses include:
- positive relationship
- relationship for slides is stronger than flows
- [justify with data from Resource 8]

(f) Explain how natural factors can result in the occurrence of mass movement in the humid tropics. [6]

One natural factor contributing to mass movement in the humid tropics is intense and prolonged rainfall. Tropical regions experience frequent, heavy rainstorms, especially during the wet season. Water infiltration into the soil increases the moisture content,

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leading to a condition called water saturation. When soil becomes saturated, its cohesion decreases, and the friction that holds particles together weakens. The added weight of the water combined with the reduced stability of the slope makes the soil or rock mass more likely to slide downslope. Continuous rainfall can also lead to the formation of pore pressure—the pressure exerted by water between soil particles—which further reduces the stability of the slope and triggers landslides.

The humid tropics often feature rugged landscapes with steep slopes due to tectonic activity, volcanic formations, and erosional processes. These steep terrains are inherently unstable, and gravity constantly acts on the materials on the slopes. In areas where the angle of the slope exceeds the natural angle of repose (the steepest angle at which material remains stable), the likelihood of mass movement increases. Slopes in the tropics, particularly those formed by volcanic activity, are often young and have not yet stabilized, making them more prone to landslides, rockfalls, and other forms of mass movement.

The type of soil and rock in tropical regions plays a critical role in the occurrence of mass movements. Many tropical soils, such as lateritic soils, are formed through intense weathering due to the hot and humid climate. These soils tend to be deep and can become very unstable when saturated with water. The presence of weak or fractured bedrock underneath also contributes to slope instability. Certain rocks, like clay-rich soils, can expand and become slippery when wet, further reducing the stability of the slope. The combination of deeply weathered soils and water-susceptible rocks increases the likelihood of slope failure during heavy rainfall events.

In the humid tropics, dense vegetation is common, and tree roots play an important role in stabilizing slopes by binding the soil together and reducing erosion. However, when vegetation is removed—due to natural events such as landslides, storms, or forest fires—the stabilizing effect of roots is lost. Even in densely vegetated areas, mass movement can still occur, particularly when soil saturation from intense rainfall reduces root cohesion. In some cases, the weight of trees and vegetation, particularly on very steep slopes, can add to the stress on the slope, making it more susceptible to failure.

Many tropical regions lie in tectonically active zones, such as those near plate boundaries or volcanic arcs. Tectonic activity, including earthquakes and volcanic eruptions, can trigger mass movement by shaking the ground, loosening soil, and causing rockfalls and landslides. Even small earthquakes can destabilize already saturated or fractured slopes, leading to large-scale movement of material downslope. In volcanic regions, the combination of steep slopes and volcanic deposits, which are often loose and unconsolidated, makes mass movements more common, especially following seismic activity.

The tropical climate promotes intense chemical weathering of rocks due to the high temperatures and constant moisture. Over time, rocks are weakened as minerals dissolve or are altered, making them more prone to breaking apart and moving downslope. The process of erosion, driven by water runoff, can also undercut slopes and remove the base material that supports higher layers, making them unstable. Weathering, combined with erosional forces, creates a natural environment conducive to mass movement in tropical regions.

Tropical cyclones, hurricanes, and storms are common in the humid tropics, bringing with them strong winds and heavy rainfall. These extreme weather events can trigger widespread mass movements by overwhelming the natural drainage capacity of the soil, causing rapid water infiltration and slope failure. The combination of intense rainfall and wind-driven erosion makes slopes vulnerable to landslides during and after such storms.

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The movement of surface water and groundwater in tropical areas contributes to mass movement by altering the balance of forces on slopes. Surface water from heavy rains can flow rapidly over slopes, eroding the surface and removing the material that supports steeper areas. Groundwater flow, particularly in karst regions where the water dissolves underlying rock, can create cavities or reduce support in the subsurface, leading to the collapse of material above and triggering landslides or sinkholes.

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Section B

Answer **either** question 3 **or** question 4 and answer **either** question 5 **or** question 6.

Cluster 1 Development, Economy and Environment

3 Evaluate the value of Malthus' view in explaining population-resource relationships today.

[20]

Thomas Malthus, an 18th-century British economist, is renowned for his views on the relationship between population growth and resources, famously expressed in his work *An Essay on the Principle of Population* (1798). Malthus argued that while population grows geometrically (exponentially), resources, particularly food supply, grow arithmetically (linearly). According to his theory, unchecked population growth would eventually outstrip food production, leading to famine, disease, and other population checks, which he termed "Malthusian crises."

Malthus' view remains valuable in highlighting the potential consequences of overpopulation, particularly in regions where population growth outpaces the capacity to produce or distribute resources. Some developing countries, especially in Africa and South Asia, experience rapid population growth, leading to strain on agricultural systems, water supplies, and basic services like healthcare and education. In these regions, Malthusian concerns about the potential for famine and poverty still resonate, as resource availability struggles to keep up with growing populations.

In areas facing environmental degradation or water scarcity, Malthus' warning about the imbalance between population growth and resource limits echoes the current challenges of sustainability and food security. For example, water shortages in countries like Yemen or Ethiopia could be seen as a modern Malthusian crisis, where resource scarcity leads to social and economic instability.

One of the limitations of Malthus' theory is his failure to anticipate the role of technological advancements in expanding the capacity to produce food and resources. The Green Revolution of the mid-20th century dramatically increased agricultural productivity through the development of high-yield crop varieties, fertilizers, irrigation techniques, and modern farming equipment. These innovations have allowed food production to keep pace with, and even exceed, population growth in many parts of the world. The development of genetically modified organisms (GMOs), hydroponics, and vertical farming are further examples of how technology continues to challenge Malthusian predictions.

Today, global food production is more than sufficient to feed the global population, and hunger is largely a result of distribution issues, inequality, and political instability rather than absolute resource limits. Thus, Malthus' prediction of widespread famine due to population growth seems less applicable in a world where technology can dramatically increase resource efficiency.

Many industrialized nations, such as Japan and parts of Europe, are experiencing population decline and aging populations, presenting economic challenges that are the opposite of those predicted by Malthus. This trend suggests that population growth is not an inevitable consequence of human reproduction and that social and economic factors can bring about a natural balance between population and resources.

4 'Multilateral institutions have limited influence on Transnational Corporations (TNCs).'

Evaluate this statement.

[20]

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While multilateral institutions, such as the United Nations (UN), World Trade Organization (WTO), International Monetary Fund (IMF), and World Bank, play important roles in setting global norms and regulations, their ability to regulate and control the behavior of TNCs is constrained. However, these institutions can still exert influence through various means. A balanced evaluation of this statement requires examining both the limitations and the ways in which multilateral institutions can influence TNCs.

One of the reasons multilateral institutions have limited influence on TNCs is the issue of sovereignty. TNCs operate in multiple countries, each with its own legal frameworks, and are often subject to national regulations rather than global ones. Multilateral institutions lack enforcement mechanisms to directly regulate TNCs because they must respect the sovereignty of nation-states. This jurisdictional fragmentation means that TNCs can navigate between different regulatory environments, often using regulatory arbitrage to exploit weaker laws in certain countries. For instance, TNCs may shift production or operations to countries with more lenient labor or environmental standards, limiting the ability of multilateral bodies to hold them accountable globally. Multilateral institutions generally rely on voluntary agreements and soft law instruments rather than binding regulations to influence TNC behavior.

Multilateral institutions are often hampered by the absence of effective global enforcement mechanisms. While some institutions, such as the WTO, have dispute settlement mechanisms that can resolve trade issues between states, there are no equivalent bodies with the power to hold TNCs accountable across borders.

While multilateral institutions may not have direct regulatory authority, they play an essential role in setting global norms and standards that influence the behavior of TNCs. Multilateral institutions like the WTO have significant influence over the trade and investment environment in which TNCs operate. The WTO's rules on trade liberalization, intellectual property, and dispute settlement affect how TNCs conduct business across borders. While these rules are designed to facilitate global trade, they also establish legal frameworks that TNCs must navigate.

Cluster 2 Tropical Environments

5 Evaluate the influence of the El Niño Southern Oscillation in affecting rainfall in the tropics.

[20]

The El Niño Southern Oscillation (ENSO) is a significant climatic phenomenon that affects global weather patterns, particularly in the tropics. ENSO is characterized by two main phases: El Niño (the warming phase) and La Niña (the cooling phase). These phases involve variations in sea surface temperatures and atmospheric pressure in the equatorial Pacific Ocean, with significant implications for rainfall distribution across tropical regions. Understanding ENSO's influence on tropical rainfall requires evaluating the mechanisms through which El Niño and La Niña impact atmospheric circulation, as well as their consequences for different regions.

During El Niño, the warming of sea surface temperatures in the central and eastern Pacific alters normal atmospheric circulation patterns. This shift weakens the trade winds that typically blow from east to west across the Pacific. As a result, warm water accumulates in the eastern Pacific, while the western Pacific experiences cooler sea surface temperatures and reduced convection. This phenomenon suppresses rainfall in the western Pacific and South Asia. Countries like Indonesia, the Philippines, and parts of India often experience drought during El Niño years because the moisture-laden air and rain-producing convection are shifted eastward.

The impact of reduced rainfall in these areas can be profound, leading to water shortages, failed crops, and increased risks of wildfires. For instance, severe droughts associated with El Niño events in the 1997–1998 and 2015–2016 periods caused significant agricultural losses in Southeast Asia, as well as damaging forest fires in Indonesia. In contrast to the dry conditions in the west, El Niño brings above-average rainfall to the eastern Pacific and parts of Central and

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South America. The shift in warm water and convection towards the central and eastern Pacific leads to enhanced rain-bearing clouds and storms over countries like Ecuador, Peru, and parts of coastal Central America. This increased rainfall can result in flooding, landslides, and other water-related hazards.

El Niño also influences the formation and distribution of tropical cyclones (hurricanes, typhoons) in the tropics. During El Niño, warmer sea surface temperatures in the eastern Pacific lead to an increase in the number and intensity of cyclones in that region. In contrast, cyclone activity tends to decrease in the Atlantic and western Pacific, where cooler sea surface temperatures and stronger wind shear inhibit storm formation. This shift in cyclone activity can influence rainfall patterns, as tropical cyclones are often responsible for delivering large amounts of rainfall in certain regions. The reduction in cyclones in the western Pacific can exacerbate drought conditions during El Niño years, while the increase in cyclone activity in the eastern Pacific can contribute to heavier rainfall and flooding.

The El Niño Southern Oscillation (ENSO) has a profound influence on tropical rainfall patterns, creating significant shifts in precipitation across different regions. These shifts in rainfall have far-reaching consequences, affecting agriculture, water resources, natural hazards, and economic livelihoods in tropical regions.

6 Evaluate the influence of climate in the formation of fluvial landforms in the humid tropics.

[20]

The formation of fluvial landforms such as braided rivers, meandering rivers, and river deltas is significantly influenced by climatic conditions, particularly in the humid tropics. These landforms result from the complex interactions between water flow, sediment transport, and the geomorphological characteristics of the landscape. Climate plays a crucial role in shaping these dynamics, and understanding its influence requires examining how precipitation patterns, temperature, and seasonality affect river behavior and landform development. Climate shapes the hydrological dynamics, sediment transport, and geomorphological processes that govern the development of these landforms. Intense rainfall, seasonal variability, vegetation cover, and climatic changes play crucial roles in determining river behaviour and landform evolution.

While climate acts as a significant driver of fluvial landform formation, it interacts with various geological, ecological, and anthropogenic factors that can further influence river dynamics. Understanding these relationships is critical for predicting how climate change may impact fluvial systems and the associated ecosystems and human communities that rely on them in the humid tropics.

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Generic Level Descriptors for H2 Essays		
Level	**Marks**	**Descriptors**
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well- supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

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